

Enterprise Procurement Intelligence Platform

Executive Summary



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Prepared For: Industry Professionals, Clients, Hiring Managers

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Classification: Public – For Portfolio and Case Studies

1. Platform Overview

The Business Challenge

Procurement organizations face a fundamental problem: they manage hundreds of suppliers and thousands of invoices, but lack integrated visibility into supplier risk, purchasing opportunities, and vendor performance. Most organizations track suppliers in silos—the Vendor Master holds risk ratings, the Accounts Payable (AP) system holds invoice errors, and the Purchasing system holds order history. Rarely are these datasets merged to answer critical questions about supplier health, contract suitability, and supplier base optimization.

The Solution

The Procurement Intelligence Platform is an integrated analytics solution that transforms raw ERP data into actionable procurement intelligence across three interconnected decision engines:

Engine	Core Question	Output
Supplier Risk Management	Which suppliers create operational risk?	Retain / Monitor / Replace
Contract Intelligence	Which products should be under contract?	Contract / Review / Phase-Out
Supplier Development	Which suppliers deserve more business?	Strategic / Preferred / Approved / Monitor

The Investment

The platform was built using existing ERP data (vendors, invoices, purchase orders) and open-source analytics tools (SQL, Tableau). No additional data infrastructure was required. The analysis spans 2023–2025 transaction history, covering 88 suppliers, 2,168 invoices, and \$627M in annual spend.

2. Why This Report Matters

Traditional procurement reporting tells leaders what happened. This report tells leaders what to do next.

Traditional Reporting	Procurement Intelligence Report
Summarizes historical spend	Prioritizes supplier intervention
Shows invoice volumes	Quantifies savings opportunities
Lists suppliers by spend	Detects hidden operational risk
Reports on payment cycles	Prevents poor contracting decisions
Provides static dashboards	Supports executive governance

This is a decision engine that answers the questions procurement leaders ask daily:

- Which suppliers should I call today?
 - Which contracts should I negotiate this quarter?
 - Which suppliers should I invest in developing?
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3. Key Findings by Engine

Engine 1: Supplier Risk Management

The Problem: Static risk labels ("Low," "Medium," "High") in the vendor master do not capture operational friction—invoice errors and payment rejections that drive AP inefficiency and delay payments.

The Approach: We developed a weighted scorecard combining Spend Exposure (30%), Invoice Error Rate (30%), Payment Rejection Rate (20%), and Inherent Risk Level (20%). Suppliers scoring ≥ 70 are flagged for Replacement, 40–69 for Monitoring, and < 40 for Retention.

The Findings:

Metric	Value
Total Vendors Analyzed	88
Total Annual Spend	\$626.9M
Vendors in Monitor Zone (40–69)	18 vendors (20%) – \$174.8M (28% of spend)
Vendors in Retain Zone (< 40)	70 vendors (80%) – \$452.1M (72% of spend)
Vendors in Replace Zone (≥ 70)	0

Critical Insights:

- Accenture PLC – Classified "Low Risk" but drives the highest invoice error rate (10.71%), 4.7 $\times$ above the 2.29% average.
- CBRE Group – 56.25% payment rejection rate, 3 $\times$ above the 18.76% average, signaling an urgent AP process review.
- Vintage Store – The sole "High Risk" vendor performs flawlessly (0% errors, 3.6% rejections), demonstrating that legacy risk labels are unreliable.

Hero Finding: "Low Risk" does not mean "Low Friction." A dynamic, transaction-based scorecard is essential for effective procurement oversight.

Engine 2: Contract Intelligence

The Problem: Thousands of products are purchased transactionally, resulting in missed volume discounts, inconsistent pricing, fragmented procurement, and high administrative overhead.

The Approach: We developed a weighted scorecard across Spend Impact (30%), Purchase Frequency (25%), Demand Predictability (20%), Price Volatility Opportunity (15%), and Supplier Concentration (10%). The model includes a Phase-Out Override to prevent contracting products with collapsed demand (YoY decline >50% AND 2025 spend <25% of 2023 spend).

The Findings:

Metric	Value
Products Analyzed	126
Strong Contract Candidates (≥ 70)	1 product – Power Tools (\$2.47M spend, \$198K savings)
Strategic Review Candidates (40–69)	50+ products
Phase-Out Flagged	35+ products
Total Addressable Savings	\$2.7M

Critical Insights:

- Power Tools is the only immediate contract candidate with \$2.47M spend, 130% YoY growth, and \$198K estimated annual savings.

- Implementation Service had \$2.69M in 2023 spend but only \$11K in 2025 (-93.8% decline)—without the trend overlay, this would have been recommended for contracting.
- The Phase-Out Override prevented erroneous contracting decisions for 35+ products that are effectively dying.

Hero Finding: Traditional aggregate analysis would have recommended contracts for dying products. The trend overlay protected procurement from signing long-term agreements for declining demand.

Engine 3: Supplier Development

The Problem: Suppliers are evaluated primarily by spend volume, with no objective framework to determine which suppliers deserve more business or require risk mitigation.

The Approach: We built a 5-dimension supplier scoring model with fixed business thresholds (not dataset-dependent normalization): Business Importance (20%), Operational Quality (30%), Internal Processing Efficiency (20%), Strategic Position (20%), and Risk (10%). Strategic intelligence flags overlay the score to drive specific actions.

The Findings:

Metric	Value
Suppliers Analyzed	12
Total Spend Under Management	\$101.03M
Preferred Suppliers (70–84)	3 suppliers – \$22.1M
Approved Suppliers (50–69)	9 suppliers – \$65.6M

Strategic Partners (≥ 85)	0
Suppliers with Concentration Risk	10 suppliers – \$87.7M

Critical Insights:

- Concentration risk is the dominant issue – 10 out of 12 suppliers have limited product diversity (≤ 2 categories), representing \$87.7M in spend.
- Vintage Store is the top-performing supplier (100% accuracy, 0% rejections, 135% growth) but carries concentration risk—requiring both "Maintain" and "Develop Backup Supplier" actions.
- Rockwell Automation is a consolidation opportunity—perfect accuracy with only \$4.6M spend.

Hero Finding: Concentration risk, not supplier quality, is the dominant issue.

Procurement should develop backup suppliers for 10 vendors representing \$87.7M in spend.

4. Integrated Decision View

When the three engines are viewed together, a unified picture of the supplier base emerges:

Supplier	Risk Engine	Contract Engine	Development Engine	Composite Action
Vintage Store	● Retain (38.5)	N/A	● Preferred (81.5)	Expand relationship + Develop backup

Michelin	 Monitor (59.8)	N/A	 Approved (65.0)	Develop backup supplier
Accenture	 Monitor (44.3)	N/A	Not in dataset	Executive review + Performance improvement
Rockwell	N/A	N/A	 Approved (68.0)	Consolidation opportunity
Power Tools	N/A	 Contract (74.4)	N/A	Negotiate contract
Implementation Service	N/A	 Phase-Out	N/A	Exit / renegotiate

5. Illustrative Annual Value Opportunity

Note: The following figures are modeled estimates derived from the analysis. Actual savings would require validation against live organizational data and supplier negotiations.

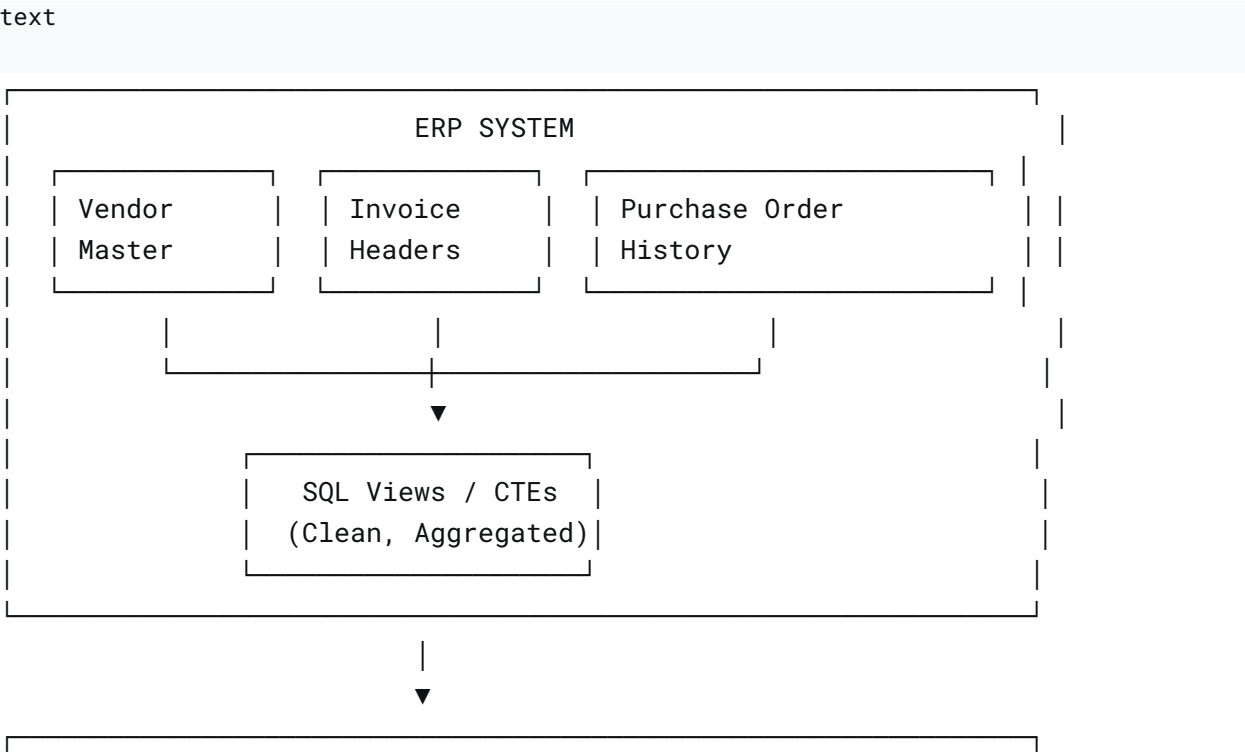
Impact Area	Illustrative Annual Value
Contract Negotiation Savings	\$2.7M

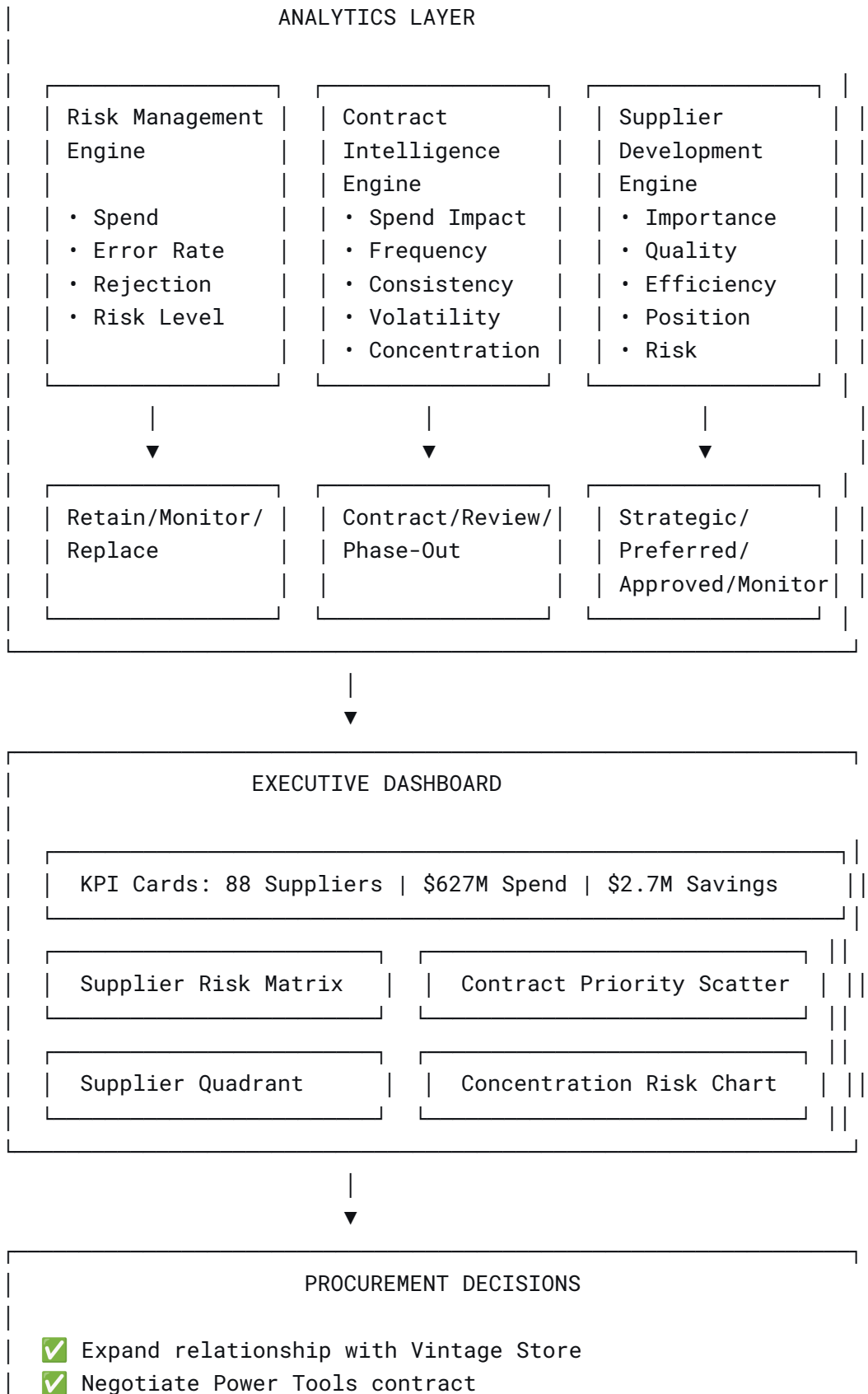
AP Friction Reduction	\$0.8M
Supplier Consolidation Savings	\$0.5M
Total Addressable Savings	\$4.0M+

Our findings was able to showed that by optimizing the supplier base and contract strategy, it was to recover an estimated \$4.0M+ annually from a small number of high-impact actions—negotiating Power Tools, developing backup suppliers for 10 concentration-risk vendors, and reducing AP friction from high-error suppliers like Accenture and Fastenal.

6. Platform Architecture

The platform follows a modern analytics architecture, transforming raw ERP data into executive decisions through a structured, repeatable pipeline.





✓	Develop backup suppliers for 10 concentration-risk vendors	
⚠	Executive review for Accenture	
⚠	Performance Improvement Plan for Fastenal	

7. Strategic Recommendations

Immediate (Next 30 Days)

Action	Owner	Rationale
Negotiate Power Tools contract	Category Manager	\$2.47M spend, 130% growth, \$198K savings
Audit Accenture's invoice template and PO matching	AP Lead + Vendor Manager	10.71% error rate—4.7× above average
Investigate CBRE Group remittance process	AP Manager	56.25% rejection rate—highest in dataset
Develop backup suppliers for 10 concentration-risk vendors	Procurement Director	\$87.7M in spend exposed to single-product dependency

Short-Term (90 Days)

- Implement Quarterly Business Reviews (QBRs) for 18 Monitor vendors.
- Schedule Strategic Reviews for 50+ products in the 40–69 score band.
- Validate concentration risk with business stakeholders.
- Establish alerting thresholds: >10% rejection rate = process audit; >5% error rate = request corrected invoices.

Long-Term (12 Months)

- Automate the scorecards – Re-run all models quarterly.
- Integrate with the ERP – Flag suppliers and products automatically when they cross thresholds.
- Expand to non-USD vendors – Apply the same methodology to EUR and GBP cohorts.

8. Implementation Roadmap

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IMPLEMENTATION ROADMAP
PHASE 1: SQL Data Model (COMPLETE)
✓ Vendor aggregation CTE ✓ Invoice aggregation (no double-counting) ✓ Product metrics ✓ Scoring logic (Risk, Contract, Supplier)
PHASE 2: Business Intelligence (NEXT 30 DAYS)
 Tableau dashboards (3 engines)  Executive KPI cards  Scatter plots, treemaps, heatmaps
PHASE 3: ERP Integration (90 DAYS)

- 🔄 Automated scoring refresh
- 🔄 Real-time data ingestion
- 🔄 Alert triggers at thresholds

PHASE 4: Automated Alerts (6 MONTHS)

- ⚠️ Email alerts for new risks
- ⚠️ Dashboard notifications

PHASE 5: Machine Learning (12 MONTHS)

- 🤖 Supplier churn prediction
- 🤖 Price volatility forecasting

PHASE 6: AI Procurement Copilot (18 MONTHS)

- 🧠 Natural language queries
- 🧠 Autonomous decision-making

9. Project Summary

The Procurement Intelligence Platform provides procurement leadership with a unified, data-driven view of supplier risk, contract opportunities, and supplier base optimization. The platform is designed to help procurement operate more proactively by providing data-driven insights into supplier risk, contract opportunities, and supplier base optimization.

Key takeaways from this project include:

1. Static risk labels are unreliable. Vintage Store (the sole "High Risk" vendor) is actually the safest supplier; Accenture ("Low Risk") drives the highest error rate.
2. Concentration risk is the dominant issue. Ten suppliers account for \$87.7M in spend with limited product diversity—backup suppliers are urgently needed.

3. Only one product (Power Tools) meets the contract threshold. Fifty products require validation; 35+ are dying and should be avoided.
4. Illustrative annual savings of \$4.0M+ are achievable through targeted contract negotiation, AP process improvement, and supplier consolidation.

The platform transforms raw ERP data into actionable intelligence, enabling procurement to move from reactive reporting to proactive decision-making.

10. Reference to Full Project Report

This Executive Summary provides a high-level overview of the Procurement Intelligence Platform. For detailed methodology, complete findings, SQL code, and dashboard specifications, please refer to the full project report:

[Download the Full Report](#): *Procurement Intelligence & Supplier Optimization Analytics Platform – Integrated Three-Case Study Report*

The full report includes:

- Complete methodology for all three decision engines
 - Detailed findings with supporting data
 - SQL code for all models
 - Tableau dashboard blueprints
 - Integrated decision analysis
 - Assumptions and limitations
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